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Tuberous sclerosis complex (diagnostic criteria)



Last revised by Rohit Sharma (/users/rohitpi/?lang=us) on 14 Jun 2026

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The **tuberous sclerosis complex diagnostic criteria** have been developed to aid the diagnosis of tuberous sclerosis complex (/articles/tuberous-sclerosis-complex/?lang=us). These were initially published in 1998 ¹ and were subsequently updated in 2012 ² and then 2021 ³ by the International Tuberous Sclerosis Complex Consensus Group. The below criteria describe the 2021 updates.

Diagnosis

Clinical criteria

- **definite tuberous sclerosis complex:** either
 - 2 major features; OR
 - 1 major and ≥ 2 minor features
- **possible tuberous sclerosis complex:** either
 - 1 major feature; OR
 - ≥ 2 minor features

Major features

- hypomelanotic macules (/articles/hypomelanotic-macules/?lang=us) (≥ 3 , ≥ 5 mm diameter)
- angiofibromas (/articles/facial-angiofibroma/?lang=us) (≥ 3) or fibrous cephalic plaque
- ungual fibromas (≥ 2)
- shagreen patch (/articles/shagreen-patch/?lang=us)
- multiple retinal hamartomas
- multiple cortical dysplasias (include tubers and cerebral white matter migration lines) (/articles/cortical-tuber)
- subependymal nodule (/articles/subependymal-nodule/?lang=us) (≥ 2)

- angiomyolipomas (/articles/angiomyolipoma?lang=us) (≥ 2)^{*}

* a combination of the two following major features (LAM and angiomyolipomas) without another feature does not meet criteria for a definite diagnosis of tuberous sclerosis complex.

Minor features

- confetti skin lesions
- dental enamel pits (≥ 3)
- intraoral fibromas (≥ 2)
- retinal achromic patch
- multiple renal cysts
- nonrenal hamartomas
- sclerotic bone lesions

Genetic criteria

The identification of either a *TSC1* or *TSC2* pathogenic mutation is sufficient to make a definite diagnosis of tuberous sclerosis complex.

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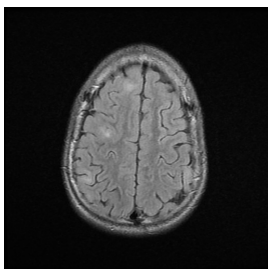
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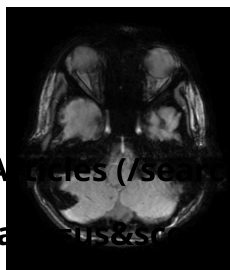
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